



Department of Information Technology

Academic Year 2025 – 2026 (Odd Semester)

Degree, Semester & Branch : V Semester-B. Tech Information Technology

Course Code & Title : IT3501 Full Stack Web Development

Name of the Faculty member (s) : S. Sakkaravarthi, AP/IT

Innovative Practice Description

- **Unit / Topic** : Unit 4 / Configuring Routes
- **Course Outcome** : CO4
- **Topic Learning Outcome:** TLO16
- **Activity Chosen** : Think Pair Share
- **Justification:**
 - The Think–Pair–Share activity is adopted to help students deeply reflect on and implement route configuration concepts in an Express.js application. Understanding how to configure routes is a key skill for building functional and responsive web services using Node.js.
- **Time Allotted for the Activity:** 20 Minutes
- **Details of the Implementation:**
 - The Faculty introduced the fundamentals of configuring routes in Express, including setting up the framework, defining route handlers, managing HTTP methods, and working with dynamic route parameters. The Think–Pair–Share activity then took place for 20 minutes.
 - Students initially used their individual time to think about the key steps in configuring routes in Express, including installing the required dependencies, setting up the project, and creating various route handlers, as illustrated in Figure 1.
 - After this individual thinking phase, students discussed their ideas and refined their route configurations with nearby peers, as shown in Figure 2.
 - In the final phase, selected students presented their route configurations to the entire class and explained how they applied routing concepts in Express.js, as depicted in Figure 3.
- **CO – PO / PSO mapping:**

CO	PO1	PO2	PO3	PO9	PO10	PSO1
CO4	3	2	3	2	2	3

(1 – Low 2 – Moderate 3 – High)

- **PO / PSO mapped:**

Innovative practice	PO1	PO2	PO3	PO9	PO10	PSO1
	3	2	3	2	2	3
Justification for correlation	Students will be able to understand the concept of Express set up, handling routes in Express.js application	Students will be able to apply Express based routing with various HTTP methods in real time applications	Students will be able to write Simple web server programs using Express.js	Students will be able to use modern development environment like Express framework to develop modern web applications	As the students participated in the activity as a team, their team work skills improved	Students gain confidence and improve their communication skills by addressing the audience during the activity.

- **Images / Screenshot of the practice:**



Fig 1: Students thinking about the given program Fig 2: Pair with neighbour and discuss answer



Fig 3: Rheyashree share the answer to entire class

- **Reflective Critique:**

- ❖ ***Feedback of practice from students and other stakeholders:***

- The activity helps students understand how to configure Express.js routes and manage route-based logic for various application needs.
 - Many of them expressed that the presentation phase helped them identify mistakes and learn different approaches from their peers.

- ❖ ***Benefit of the practice:***

- Students felt more assured when organizing and writing route handlers in Express.
 - The activity created an environment where even reserved students contributed more actively.

- ❖ ***Challenges faced in implementation:***

- More time was needed for students to fully share their thoughts and perspectives.
 - A few students lacked basic knowledge of MongoDB, which made collection management difficult for them.

In the next class, the instructor explained how to organize an Express project by separating routes into different modules and demonstrated a simple folder structure to make the concept easier for students to follow.

References:

- https://www.ritrjpm.ac.in/images/computer-science/2022-2023/1_VA_CS8603_Learning_teaching.pdf
- https://www.ritrjpm.ac.in/images/computer-science/2022-2023/6_GS_CS8592_learning_by_teaching.pdf

Signature of Faculty Member

HOD